Literature Review and Plan

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1 Introduction

Since the Dawn of Time, humankind has lusted after knowledge, but has been hamstrung by its inability to accurately judge uncertainty. This is essential, since it is only by quantifying our ignorance that we can know where to search for illumination.

The modern world is awash with data, however there is not a direct relationship between the volume of data and the volume of information. This was clearly demonstrated by a lecturer in my undergraduate course who showed a YouTube video of a karaoke performance, remarking that 'Google has served us several million bits of data, yet the total useful information in this room has decreased'.

Luckily, in tandem with this massive rise in the volume of data, the process of creating and analysing the data has been federated and democratised. The abundance of fast network connections, powerful computers and an interested populace has created Citizen Science. There are projects that use spare compute cycles to fold proteins[1], spare brain power and time to classify astronomical objects[2] or spare money to buy small radiation detectors to provide a more accurate radiation report in Japan[3].

Any project that relies on data from volunteers working with commodity equipment must be capable of assessing and dealing with the reliability of the actors and the infor-

[1] http://folding.stanford.edu
[2] http://www.galaxyzoo.org
[3] http://map.safecast.org/map:full/140.0,38.0,6

mation they provide. A Bayesian framework allows us to express our beliefs about the reliability of the agents using the language of probability theory.

In this DPhil, I will be investigating ways to model beliefs and uncertainties about the world, and direct agents based on these beliefs. This direction may take the form of danger avoidance, for example by routing people around radiation, or information gathering to improve our knowledge of an object or region of interest.

The literature review focuses on the history of using noisy inputs for regression, which is the subject of the companion paper. It also provides a short overview of Gaussian Processes (GPs), and methods that use the properties of the GP to explore the world or a function of interest.

The plan section consists of an overview of the hardware platform the DPhil will be built on, followed by a series of projects culminating in papers. These projects focus on several aspects of crowd sourced data gathering, particularly realistic assessments of the quality of the data and directing participants to the best locations to gather the data.

2 Literature Review

2.1 Gaussian Processes

The GP is a stochastic process which is a prior over functions $f(\mathbf{x})$,

$$f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), C(\mathbf{x}, \mathbf{x}')). \tag{1}$$

Gaussian noise corrupted observations of this function are given by $\mathbf{y}$,

$$\mathbf{y} = f(\mathbf{x}) + \epsilon. \tag{2}$$

The Gaussian Process is specified by its mean and covariance functions. The mean function $m(\mathbf{x})$ maps the input variables $x \in \mathcal{X}$ to $\mathbb{R}$, while the covariance function $k(\mathbf{x}, \mathbf{x}')$ maps each of $x \in \mathcal{X}$, $x' \in \mathcal{X}$ to $\mathbb{R}$. Given a set of input points we can com-

pute the Gram matrix K whose entries are $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$. To be a valid covariance, the Gram matrix must be positive semidefinite (PSD).

A common choice for the covariance function is the Exponentiated Quadratic (EQ) kernel with Automatic Relevance Detection (ARD),

$$C(\mathbf{x}, \mathbf{x}') = s^2 \exp\left(-\frac{1}{2}\sum_{i=0}^{D}\frac{(x_i - x'_i)^2}{l_i^2}\right). \tag{3}$$

In (3), each input dimension has length-scale l_i, which is used to implement *shrinkage*. A dimension with a short length scale has much greater effect on the output of the function, so the process can disregard irrelevant inputs by increasing their length scale.

Usually, we optimise the parameters of the kernel, s and the D dimensional vector $\mathbf{l}$ using maximum likelihood. This is optimising the parameters of the prior, and is known as Type-2 Maximum Likelihood. For a fully Bayesian treatment the hyperparameters should be marginalised, however this is usually intractable and needs to be approximated. In certain cases we can perform this marginalisation analytically. The Rational Quadratic (RQ) kernel is an example of this, where the inverse squared length scale parameters of the EQ kernel are marginalised over a gamma distribution [9].

For observed data $\mathbf{y}$ at $\mathbf{x}$, we can make predictions for unseen function values $\mathbf{f}(x_*)$ (denoted by f_* henceforth) at prediction locations $\mathbf{x}_*$ using the marginalisation properties of the multidimensional Gaussian distribution. In the following equations, subscripts are used for brevity to represent the vectors the operations are performed on. The subscript D is used to show measured data. For example, $K_{D*} = \text{Cov}(\mathbf{x}, \mathbf{x}_*)$. The joint distribution for f_* and observed values y is,

$$p\left(\begin{bmatrix}\mathbf{y}\\ \mathbf{f}_*\end{bmatrix}\right) = \mathcal{N}\left(\begin{bmatrix}\mathbf{y}\\ \mathbf{f}_*\end{bmatrix}; \begin{bmatrix}\mu_D\\ \mu_*\end{bmatrix}, \begin{bmatrix}K_{DD} + \mathbf{I}\epsilon & K_{D*}\\ K_{*D} & K_{**}\end{bmatrix}\right) \tag{4}$$

which can be rearranged to find the conditional distribution for $\mathbf{f}_*$ using the identities in Bishop [2], Appendix A applied to (4), giving

$$p(\mathbf{f}_* \mid \mathbf{x}_*,\ \mathbf{x},\ \mathbf{y}) = \mathcal{N}\left(\mathbf{f}_* \mid \mu_p,\ K_p\right) \tag{5}$$

$$\mu_p = \mu_* + K_{*D}(K_{DD} + \mathbf{I}\epsilon)^{-1}(\mathbf{y} - \mu_D) \tag{6}$$

$$K_p = K_{**} - K_{*D}(K_{DD} + \mathbf{I}\epsilon)^{-1}K_{D*}. \tag{7}$$

The posterior mean (6) of the normal distribution (5) is the most likely value of the function at the location $\mathbf{x}_*$. The mean function can be seen as a weighted sum of observations the observations y, where the weights are the relevance matrix $K_{*D}K_{DD}^{-1}$. In the case of covariance functions such as (3), more weight is given to local data than distant. A covariance function is valid for any $\mathbb{R} \rightarrow \mathbb{R}$ warp of the inputs x, allowing periodic behaviour by replacing all occurences of x with $\sin(x)$.

The posterior covariance (7) represents our uncertainty about the predictions. The diagonal of this matrix is the variance of each prediction around the mean.

2.2 World Exploration

When we are attempting to find out more about the real world, we are often constrained by physical and temporal limitations. We want to sample in the minimum number of places, and only where we think important information will be. This means that we must learn using a model that keeps track of our uncertainty and a utility (or conversely, loss) function that represents our ambitions.

In [5], the authors use the GP to assist with active sampling of a river. In this problem, the hyperparameters of the GP are not known. A measurement is taken, the hyperparameters are updated and the best location for the next measurement is computed using the new model. The authors show that this sequential system performs better than choosing sample locations *a priori*, when the model hyperparameters are not initially known. This system reduces uncertainty about the world and the system's hyperparameters at the same time.

Osborne [7] explores the idea of using the GP as a proxy for the minimisation of a difficult to evaluate function. Rather than trying to purely reduce uncertainty, here the intention is to find the global minima in the function space. Unlike [5], an *a priori* approach cannot work here even if we knew the parameters.

The algorithm proceeds by defining a loss function that we wish to minimise. In this case, we wish to find the location x_m of the function minimum $y(x_m)$. Since we want to minimise $y(x)$, the loss function $\lambda(x)$ for the minimisation is simply $y(x)$.

If we have made several observations of the function and are allowed only one more, the best place to make the observation is that with the minimum expected loss. Denoting observations as $(\mathbf{x}_0, \mathbf{y}_0)$ and defining $\eta \triangleq \min \mathbf{y}_0$, the loss of observing y at point x is

$$\lambda(y) \triangleq \begin{cases} y & \text{if } y < \eta \\ \eta & \text{if } y \geq \eta. \end{cases} \tag{8}$$

The GP gives us a distribution over y, and we can marginalise this loss over the distribution to give us an expression for the expected minimum after the next evaluation. The task now is to minimise the expected loss, which is a considerably easier problem than the original.

2.3 Noisy Inputs for Regression

Most regression systems rely on the independent variables (in this article, x) being noiseless. In general this is a flawed assumption, because the inputs must be measured using a noisy sensor.

Noisy inputs in a linear function serve to increase the apparent output noise in steep regions of the function. To visualise this, consider a linear function $f = \alpha x$. When α is 0, the function is flat. A change in the actual location of the measurement will have no effect because $f = 0$ for all possible values of x. Increasing α will increase the effect a change in x will have on the value of f. In the simple linear case explored here, we can simply

project the uncertainty in x through a linear transform to give additional uncertainty in y:

$$x \sim \mathcal{N}(\mu\, \sigma^2) \tag{9}$$

$$f = \alpha x \tag{10}$$

$$f \sim \mathcal{N}(\alpha\mu, \alpha^2\sigma^2) \tag{11}$$

In a nonlinear function, there may be other factors that affect the uncertainty in a different ways, so a more rigorous formulation is required. The problem however is clear: we underestimate variance, and give too much weight to samples taken on steep parts of function relative to shallow. If we are required to make a decision based on the result of the regression, underestimating the uncertainty may lead to making sub-optimal decisions.

For further analysis, we must declare the model:

$$x \sim \mathcal{N}(\hat{x},\ \Sigma_x) \tag{12}$$

$$f = g(\hat{x}) \tag{13}$$

$$y \sim \mathcal{N}(g(\hat{x}),\ \sigma_y^2). \tag{14}$$

We observe y and x. The true function value f is latent, as are the true sample locations $\hat{x}$.

It is illuminating to first consider modifications to least-squares regression that allow most-probable results when regressing with noise corrupted x and y. Adcock, in his 1877 Notes on The Method of Least Squares [1] asserts that the most probable line is that which minimizes the sum of the squares of the *normal* of the distance of each variable to the line. This is in contrast to the more usual least-squares fit, which minimises the sum of squares of the dependent variable to the line. Other authors, such as Wald [12] in 1940, contest that the choice of minimising the normal deviation rather than some other distance is arbitrary. Wald points out that under a transformation of the coordinate system, a line

fit under the normal deviation is not invariant.

Wald develops a method to estimate the confidence intervals and expectations of the parameters α and β of the latent function $f = \alpha x + \beta$ and the standard deviations η and ϵ of the errors corrupting the latent variables x and f. He does not, however, provide a distribution over the predictive values of y.

Wald's analysis of the identifiability of the variables and standard deivations was expanded by Reiersøl in 1950 however, as noted by Z. Griliches and V. Ringstad in their 1970 paper Error-in-the-Variables Bias in Nonlinear Contexts [4], there is limited applicability of these methods to non-linear scenarios, which are of primary interest to our investigations.

Non-parametric regression methods allow increasing model complexity with increasing amounts of data. Fan and Truong [3] considered the effects of input noise on a non-parametric kernel estimator, and developed the following estimator:

$$\hat{m}(x) = \frac{\sum_i K_n\left(\frac{x - x_{ni}}{h_n}\right) y_{ni}}{\sum_j K_n\left(\frac{x - x_{nj}}{h_n}\right)} \tag{15}$$

with

$$K_n(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-jtx} \frac{\phi_k(t)}{\phi_e(t/h_n)}\, dt \tag{16}$$

where $\phi_k(t)$ is the Fourier transform of some kernel $K(x)$ and $\phi_e(t)$ is the Fourier transform of the noise distribution. A highly peaked (and therefore certain) function will have a large bandwidth - the noise function on the denominator will have the effect of reducing the kernel's bandwidth and hence certainty in regions of high noise. In (15) this has the effect of giving observations taken in regions with large input noise.

For applications to active sampling, we are interested in finding the full predictive posterior, which has been lacking in methods explored above.

The GP provides us with a posterior predictive distribution given noiseless independent variables x, however extending this to inputs distributed as $p(x \mid \hat{x})$ around a mean $\hat{x}$ requires evaluation of the integral

$$p(f(x_*) \mid \ y, \ \theta) = \int_{-\infty}^{\infty} p(f(x_*) \mid x, \ y, \ \theta) p(x \mid \hat{x}) \, dx \tag{17}$$

which is generally intractable under the GP model due to non-linearities in the mean and covariance functions.

A immediately obvious method to approximate this integral is via Monte-Carlo, which approximate the integral operator as a sum of the integrand at N independent draws x_n from $p(x \mid \ \hat{x})$,

$$p(f(x_*) \mid y, \ \theta) \approx \frac{1}{N} \sum_{n=1}^{N} p(f(x_*) \mid \ x_n, \ y, \ \theta). \tag{18}$$

While Monto Carlo is accurate if correctly performed it requires thousands of samples to provide a reasonable approximation. Equivalently, the computation time required is thousands of times the amount required for a simple GP. For that reason we generally try to use less accurate but much faster approximations.

McHutchon's Noisy Input Gaussian Process (NIGP) and Wright's 1998 paper on Neural Network regresson with input noise [13] provide very similair solutions to the problem in (17). The authors take a Laplace approximation to the integrand, and in the limit where the input noise error is small and Gaussian an additional term is added to the output noise variance, equivalent to σ_x^2 multiplied by the gradient of the mean function. This is a clear analogy to the simple linear case shown in (11).

In both methods, the predictive mean is affected by the additional output noise. By considering the NIGP's modified mean function,

$$m(x_*) = K_{*D} \left(K_{DD} + \mathbf{I}\sigma_y^2 + \text{diag}\{\Delta_{\bar{f}} \Sigma_x \Delta_{\bar{f}}^T\} \right)^{-1} \mathbf{y} \tag{19}$$

we can see that the inflated covariance matrix $K_{DD} + \mathbf{I}\sigma_y^2 + \text{diag}\{\Delta_{\bar{f}} \Sigma_x \Delta_{\bar{f}}^T\}$ serves to reduce the weight given to measurements taken at steep sections of the function, analogous to (15).

In the Marginal Gaussian Process (MGP) developed in the companion paper to this

literature review and by Osborne *et al.* [8], we moment match the GP posterior with a Gaussian and marginalise over this new distribution. This approximation allows us to take into account the gradient of the variance with respect to the inputs and more accurately approximate the posterior variance. Extending this approximation to allow learning the input noise is one part of my intended future work.

3 Work plan

There have been two focuses to my work this year. The first has been in understanding the GP used for regression and adapting it for use in situations where there is input noise. The second has been building my skillset in scientific computing and data gathering applications on mobile phones.

Before I discuss my future work, I would like to mention some of the hardware I have been working with. This is neither future work nor a review of relevant literature, however it will be helpful for the reader to be aware of the hardware that will underpin the DPhil.

3.1 Hardware for machine learning

Real-world datasets for Machine Learning are very important - without them, we would be locked in the onanistic world of in-model tests.

I have spent some time this year developing applications to run on Android smartphones to perform some data gathering tasks. The most successful at the moment is a program to monitor chicken welfare using optical flow, working with some of the authors of [6].

Other programs tested the ability of the phones to perform signal signal processing and machine learning tasks. The insatiable demand for more accurate simulations of flying pigs has paid dividends in terms of mobile processing power. The Nexus 4, a phone available for less than £160 has a quad core 1.5 GHz processor and 2 GB of RAM. This is more than enough processing power for even rich and complex models like the GP

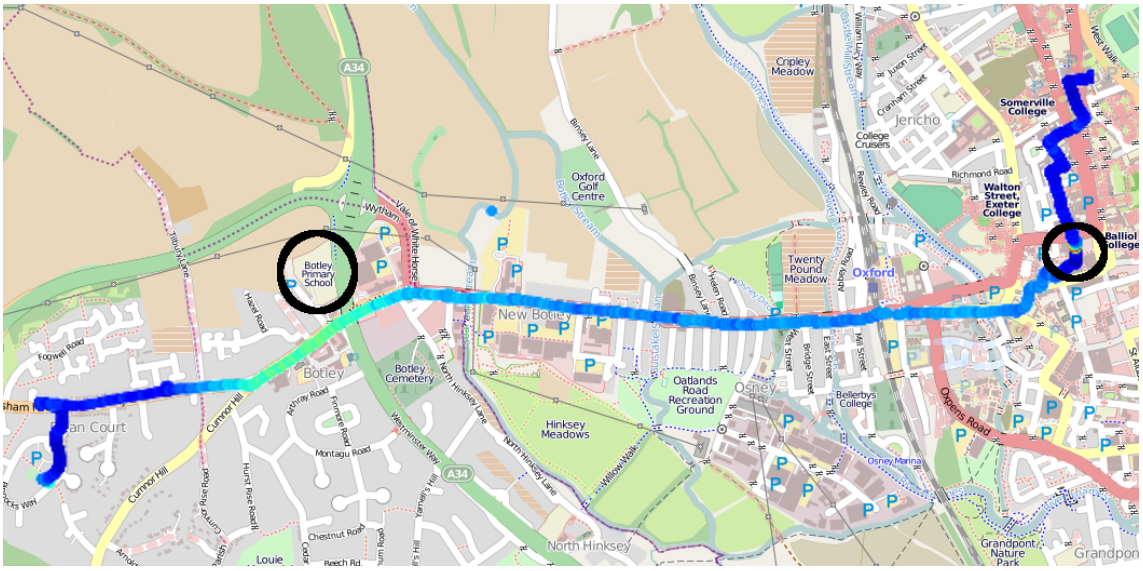

Figure 1: A map of west Oxford with CO readings from the Sensordrone. Blue denotes 0 PPM CO, shading through green to red (15 PPM).

described above.

We took advantage of the phone's bluetooth connectivity by acquiring a SensorDrone. This is a battery powered sensor capable of measuring: Carbon Monoxide (CO), reducing gasses, oxidising gasses, remote temperature, case temperature, color, pressure, proximity (capacitance) and other external sensors using its serial connection and Analogue to Digital Converter (ADC).

Currently, I am developing an application for this that records air quality indicators on my journey into the lab. I have already acquired some preliminary data, shown in Figure 1.

There are two interesting things to be seen from this. Firstly, Botley Primary School is directly under the A34 and therefore has a high concentration of CO. Additionally, the bus station in the center of town has a very high concentration of CO compared to the surrounding area. The bus station and school are both circled in black on the map.

The SensorDrone has many functions, however it sacrifices some accuracy for battery life and portability. Similarly, the measurements it makes are not independent measurements, rather a convolution of the sensors' response functions with the scalar field being

measured. Taking these factors into account is imperative.

3.2 Noisy inputs

The paper submitted along with this review shows some positive results for using the MGP to learn errors-in-variables GPs, however for real data the method we used to learn in the paper was problematic. In development of the paper we found that despite received wisdom, ARD methods in GPs rarely found a better solution than isotropic covariance functions and in general had a much larger spread in predictive log likelihood. A likely explanation is that ARD makes learning via Maximum Likelihood Estimation (MLE) difficult due to the increased chance of getting stuck in local maxima, however this is an avenue for future research.

Learning input noise adds an additional D hyperparameters. Under an EQ ARD kernel, this means that we have $2D + 2$ hyperparameters to learn. In the 7 dimensional Yacht dataset, we found that the NIGP and MGP performed better than both hetero- and homoscedastic systems. In the house prices both the MGP and the NIGP were beaten by isotropic RQ and EQ based GPs, both of which had higher means and less variability in their predictive likelihood.

The first part of my work in the coming year will be further development of the MGP, for submission to ICML in October. The main addition to the paper will be learning with the MGP, along with testing versions of the MGP and NIGP with isotropic input noise. We will carefully investigate the problems faced with ARD learning, which were noted but not investigated in depth by the companion paper.

3.3 Route selection

For crowdsourcing the gathering of data, it is important to remember that the participants may not be amenable to going very far out of their way or putting up with much inconvenience. Similairly, in a disaster response scenario, there is a balance to be struck between gathering data to help response planning and actually performing the response.

For these reasons, a large part of my work for the next year will be on using expected information to select routes from a finite set. These could be a set of routes to work, or a set of allowable paths in a multi-agent environment to avoid collisions and a radiation spill.

There is work on-going under the ORCHID project where active sampling is used to direct agents to learn about their environment, while avoiding areas likely to be dangerous. In the current version of the demo (by Steve Reece) the sample location is chosen to be at the optimal location in a manner similar to Gaussian Process Global Optimisation (GPGO), not taking the agent's dynamics into account.

Another realisation of these ideas is to generate a route based on our beliefs about the function. If we have a function in space over a road network, and a utility function that we wish to optimise, we can weight each segment of the network based on that utility function. An optimal route can then be selected using standard routing algorithms, such as A^*.

A project that is starting up based on this idea is to take the Fukushima radiation map developed by Venanzi *et al.* [11] and plan routes which give minimal radiation exposure. Venanzi's system models the radiation dose rate using a GP. In order to perform the weighting for the routing algorithm, we must integrate the GP over the route, taking into account journey time along each leg. We can exploit the structure of model by making a local linearisation to the road and analytically integrating each section using Bayesian Quadrature (BQ).

Returning to the crowd sourced data gathering, consider a scenario where we want to map the pollution in a city. Passing through quickly does not pose a health risk so we do not need to concern ourselves with agent health.

We are not simply interested in reducing uncertainty, instead we wish to find high levels of pollution, since these are the dangerous areas. This leads us to use a utility function similar to that used in GPGO, which balances exploring regions where we are ignorant and reducing already low ignorance in regions with a high expected mean. Regions where

we are highly certain of low pollution need not be explored again until the low pollution readings are so far in the past that our uncertainty has grown.

This work will follow on from the MGP work, which MGP work will take up most of September. The first deadline for International Conference on Machine Learning (ICML) is October 4. If we miss this, we will aim to submit by January 31, 2014. The route finding work will begin in October, whether or not the MGP work is finished.

The work will consist of initially generating a toy world with roads acting as pollution sources and refining the method of integration over the GP. Next I will test for a single agent in the model whether this route-based active sampling is more effective than either random selection of routes or simply taking the same route repeatedly, and if so how much. We will begin testing this active sampling in the real world using members of the lab provided with Android smart phones and a SensorDrone.

3.4 Active Sampling for Biodiversity

In concert with the New Forest Cicada Project and Timos Papadopoulos I am working on making an Android based platform to direct groups of people to find out about biodiversity and the soundscape of the world. We will use the phone's microphone to sample the soundscape and infer biodiversity from those readings. My responsibility is the development of the routing framework to direct the walker to the optimal location within the region of interest, while ensuring they stick to the paths and get back on time. These constrained routes make it possible to build on the route selection work mentioned above.

3.5 Reliability of Agents

Any crowd sourcing project must take into account the reliability of the participants. Simpson *et al.*'s work [10] provides a Variational Bayes (VB) algorithm to track and adapt to an agent's changing reliability. It has recently been extended by a member of the group, Steve Reece, to decentralise the VB update equations.

This is an important future step for any citizen science data gathering project. Following the active routing work, I will investigate coordinating the route selection of multiple human agents using these decentralised updates. This is clearly an important area of work, because we do not want to duplicate effort or ask unreliable participants to follow an unusual route. I predict that this work will begin around the end of the second year of my DPhil, however it is hard to be specific about work dependant on so many variables.

4 Risk Points

Some possible risk points:

We have an almost immediate deadline on 4 October for an ICML submission. There is a second submission date on 31 January 2014, however missing the October deadline will slow down work on the Route Decisions task. Therefore, I have left some slack in the timing for these.

The project on Active Sampling for Biodiversity has the same problem that all other projects with multiple stakeholders have, which is synchronising everyone involved. I have given a long lead time for this project, to allow for work on route selection to proceed and to allow the group time to organise.

The other risk point is the extension of Simpson's work on multiple independent classifiers to agent based data gathering. It is relies on adapting the system to a yet to be developed method, and as such any prediction is hard to make. To mitigate this risk, work will be carried out throughout the other projects to increase my understanding of and familiarity with Simpson's model.

Figure 2 shows a Gantt chart with projected time frames for each project in the DPhil. I have allowed an uninterrupted 6 months for the write up, and have placed the submission deadline 6 months early to allow for unforeseen events.

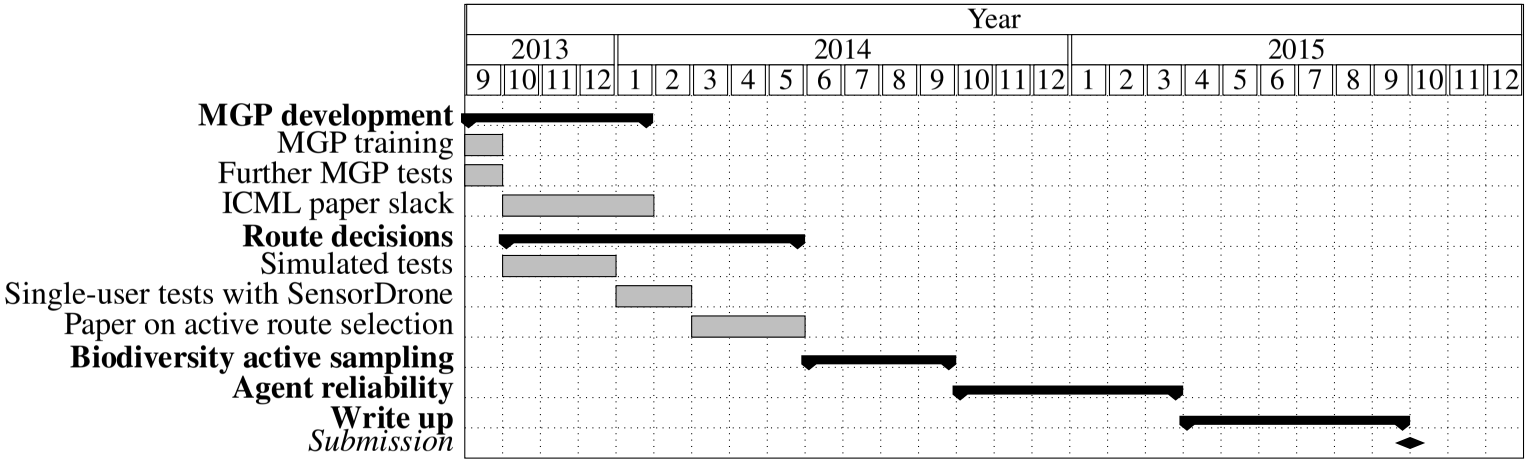

Figure 2: Gantt chart for DPhil completion.

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